# Eigen-Séta white paper: study report

Evidence for `crypto:papers/eigen-seta-whitepaper/`. Measured on a shared 4-vCPU Xeon @ 2.10 GHz VM (L0); security levels derived.

## Timings (all cores, medians)

| set | keygen | encaps | decaps | pk = ct | sk |
|---|---|---|---|---|---|
| L1 | 4.6 s | 0.20 s | 1.20 s | 1344 B | 1855 B |
| L3 | 84 s | 1.1 s | 5.8 s | 1992 B | 2730 B |
| L5 | 51 s | 2.3 s | 12.8 s | 2632 B | 3594 B |
| Q3 | 14 min (4 jobs) | 5.9 s | 37 s | 3904 B | 5311 B |
| Q5 | 33 min (4 jobs, 1 key) | 13.7 s | 89 s | 5192 B | 7050 B |

Run run-20261017T2022Z-kem-bench, eigenx `d89d790`; every result checked. The attacker's chain test costs 618–670 $F_{p^2}$ multiplication-equivalents per step by wall time, which includes setup and additions; exact counts give 188.9 $F_{p^2}$ multiplications per theta step (aburan28/eigenx#7), below the model's 256, which therefore overcharged the attacker (corrected to $2^{7.5}$).

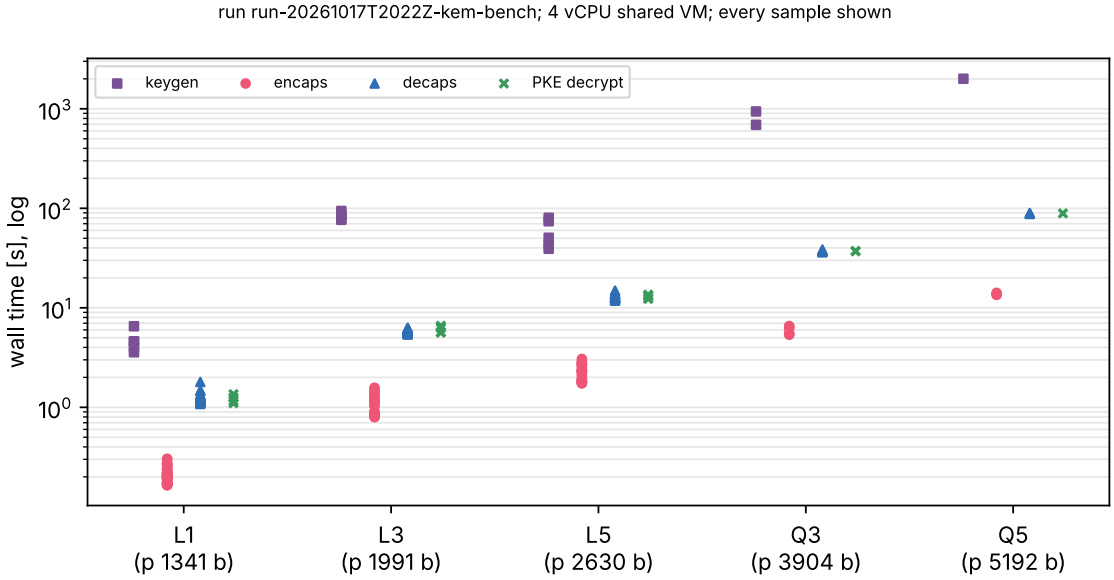



Figure 1: Every sample, log scale; measured.

## Same-host comparison

Decapsulation at level 1: ML-KEM-512 14 µs, McEliece-348864 158 µs, FrodoKEM-640 441 µs, BIKE-L1 458 µs, HQC-128 (portable) 5.1 ms, Eigen-Séta L1 1.2 s, about 89,000 times ML-KEM. liboqs 0.13.0, one pinned core, two repetitions.

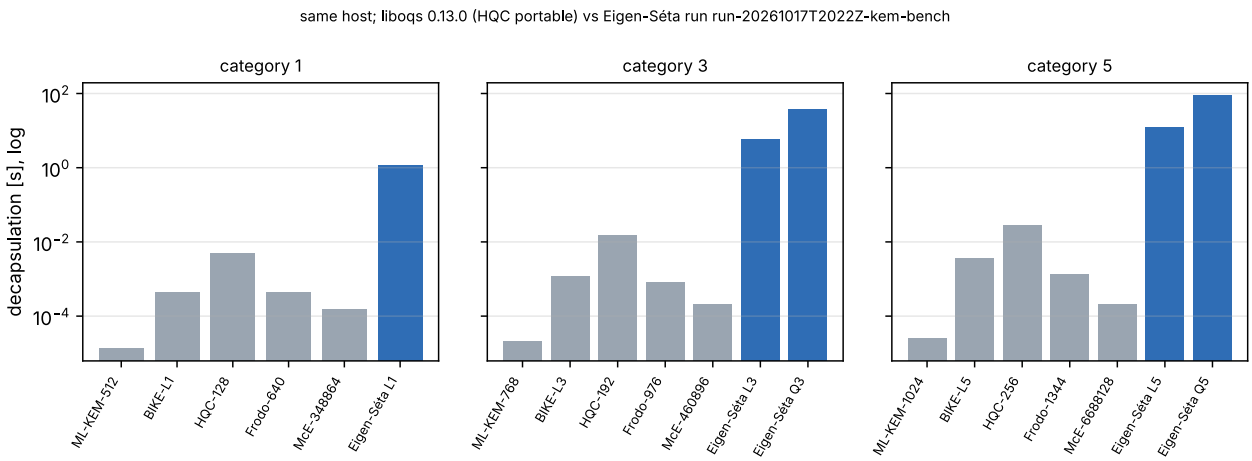



Figure 2: Decapsulation by category, log scale; measured.

## Security (derived)

L1, L3 and L5 meet categories 1, 3 and 5 only in gate count under MAXDEPTH. At least 2^λ quantum queries holds only for L5 (λ = 128), Q3 and Q5. Largest risk: the hidden orientation becoming effective, which would break every set quantumly.

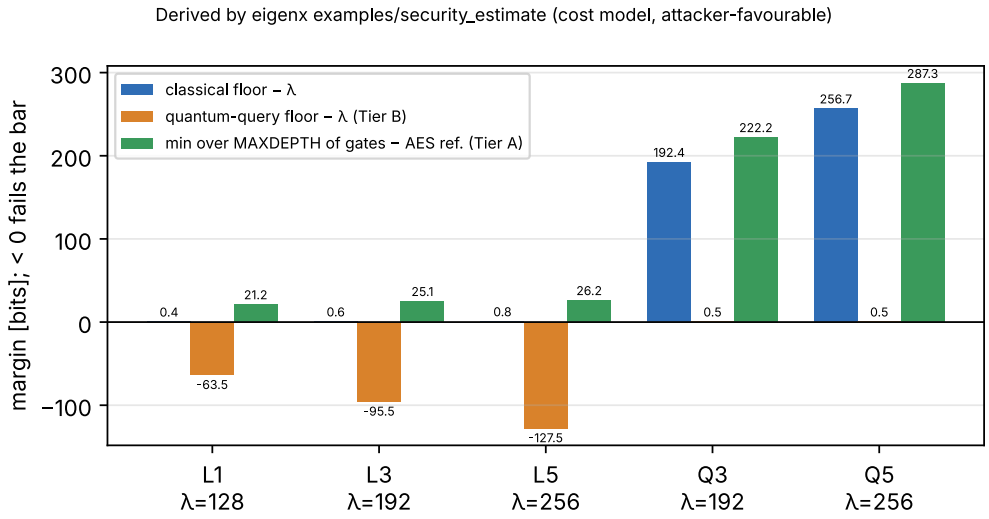



Figure 3: Margins over λ by tier; derived by the eigenx cost model.